

SCIENCE
Half Yearly Examination, 2024-25
(Sample Question Paper)

GRADE: VI

Maximum Marks: 80

Please check that this question paper contains **3** printed pages.

Answers to this Paper must be written on the paper provided separately.

Reading time: 15 minutes

Writing time: 3 hours

General Instructions:

- This question paper consists of 39 questions in 5 sections.
- All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of these questions.
- **Section A** consists of 20 objective type questions carrying 1 mark each.
- **Section B** consists of 6 Very Short questions carrying 02 marks each. Answers to these questions should be in the range of 30 to 50 words.
- **Section C** consists of 7 Short Answer type questions carrying 03 marks each. Answers to these questions should be in the range of 50 to 80 words
- **Section D** consists of 3 Long Answer type questions carrying 05 marks each. Answers to these questions should be in the range of 80 to 120 words.
- **Section E** consists of 3 source-based/case-based units of assessment of 04 marks each with sub-parts.

SECTION - A

Select and write one most appropriate option out of the four options given for each of the questions 1 – 20

Q. No.	Questions	Marks
1.	The image formed in a mirror is a. laterally inverted b. same size as of the object c. coloured d. all of these	[1]
2.	The process of conversion of water to ice is known as _____ a. evaporation b. boiling c. melting d. freezing	[1]
3.	Which of the following has the lowest content of carbohydrates? a. Banana b. Papaya c. Peas d. Potato	[1]
4.	A material through which light cannot pass at all is a. transparent b. translucent c. opaque d. insoluble	[1]

5.	Which of the following is a slow change? a. Tearing of paper b. Burning of matchstick c. Growth of a plant d. Bursting of a balloon	[1]
6.	Iodine starch test is used to test the presence of _____ in foods a. Starch b. Proteins c. Vitamins d. Fats	[1]
7.	Light travels in a _____ line. a. random b. curvilinear c. irregular d. straight	[1]
8.	Which of the following is an irreversible change? a. Breaking of an egg b. Cooking of rice c. None of the above d. Both a and b	[1]
9.	_____ is caused due to the deficiency of vitamin B12 a. Night blindness b. Scurvy c. Beriberi d. Anaemia	[1]
10.	The motion of the butterfly is_____. a. curvilinear b. rectilinear c. periodic d. circular	[1]
11.	What is an example of a common miscible liquid used in everyday life? a. Orange juice and water b. Water and oil c. Sand and Vinegar d. Milk and Sugar	[1]
12.	Choose the odd one out a. Hemp b. Ramie c. Linen d. Terylene	[1]
13.	SI stands for International _____ of Unit. a. System b. Specific c. Substance d. Simplification	[1]

14.	Which of the following materials is a good conductor of heat? a. Plastic a. Glass b. Metal c. Wood	[1]
15.	The longitudinal threads in weaving are called_____ a. Warp b. Weft c. Loom d. Spin	[1]
16.	The act of determining, size, capacity or quantity of an object is called a. units b. measurement c. pace d. motion	[1]
17.	Which of the following materials is commonly used for making windows because it is transparent? a. Wood b. Plastic c. Metal d. Glass	[1]
18.	Cotton bale covers are made of a. Coir b. Jute c. Cotton d. Linen	[1]
19.	10 millimetres is equal to a. 1 metre b. 1 centimetre c. 1 kilometre d. 1 cubit	[1]
20.	Deficiency of _____mineral causes dental decay a. Iodine b. Calcium c. Fluorine d. Phosphorus	[1]

SECTION – B

Q. No. 21 to 26 are very short answer questions.

21.	If you watch a pendulum swinging back and forth, what kind of motion is it showing? Give another example of that motion.	[2]
22.	What is the main purpose of grouping materials? Provide one example of how materials are grouped.	[2]
23.	Name the carbohydrate present in dietary fibres. Give 2 examples of food that contain dietary fibres.	[2]
24.	Write down four man-made sources of light.	[2]

25.	Is drying of clothes considered a chemical change? Explain.	[2]
26.	What is the significance of wearing clothes?	[2]
SECTION - C		
Q. No. 27 to 33 are short answer questions.		
27.	Distinguish between the image formed by a mirror and the shadow of an object. (give 3 points of each)	[3]
28.	Growth of human body is an irreversible change. Comment	[3]
29.	Water plays a significant role in our body. Illustrate.	[3]
30.	What is periodic and non-periodic motion? Give 1 example of each.	[3]
31.	Why does cooking oil float on water? Give few examples of substances that do not float on water and Why?	[3]
32.	What are synthetic fibres? Give 2 examples.	[3]
33.	a. Deficiency diseases do not spread. Why? Give 2 examples of deficiency disease. b. Obesity is unhealthy. Explain?	[3]
SECTION - D		
Q. No. 34 to 36 are Long answer questions.		
34.	Imagine you are in a science class where your teacher has brought three different types of materials for an experiment on how light behaves. The first material is a solid wooden block, the second is a piece of frosted glass, and the third is a clear plastic sheet. a. What do you think will happen when the flashlight shines on the wooden block? Can you see any light coming from behind it? b. How about when the flashlight shines on the frosted glass? Will you be able to see anything clearly on the other side? c. Finally, what happens when the flashlight shines on the clear plastic sheet? Can you see right through it? d. Which object will make a shadow and what are the three conditions required for formation of a shadow?	[1] [1] [1] [2]
35.	The sowing of seed is a physical change whereas germination of seed is a chemical change. Explain.	[5]
36.	Cotton and Jute and natural sources of fibres that are processed to form fabric. Answer the following questions to elaborate your understanding of the process. a. Which part of the jute plant produces jute? b. Name the soil type on which cotton plant is usually grown? i. Black soil ii. Clayey soil c. Briefly explain ginning of cotton? d. Explain the process of retting in jute. e. Why do we avoid wearing nylon clothes in summer and prefer wearing cotton?	[5] [1] [1] [1] [1] [1]
SECTION - E		
Q. No. 37 to 39 are case - based/data -based questions with 2 to 3 short sub - parts. Internal choice is provided in one of these sub-parts.		
37.	The Roller Coaster Ride	

	<i>Case Study:</i> At the amusement park, there is a roller coaster ride that moves in different ways. The roller coaster starts at the top of a tall hill and goes down quickly in a straight line, then moves up and down few times, and finally circles around a track. - Identify the type of motion when the roller coaster is moving down the tall hill. - Identify the type of motion when the roller coaster moves up and down. - Identify the type of motion when the roller coaster circles around the track. - If the roller coaster had to travel through a tunnel that changes direction suddenly, what type of motion would this involve? Or - What happens to the speed of the roller coaster as it moves down the tall hill? will it increase or decrease?	[1] [1] [1] [1] [1]
38.	Imagine you are cooking soup on the stove. You place a metal spoon into the hot soup to stir it. After a few minutes, you notice that the handle of the spoon, which is outside the soup, becomes warm. This situation provides an opportunity to explore the concepts of heat and electricity conduction. - How does the metal spoon conduct heat? - Why do you think the handle of the spoon gets warm even though it's not in the soup? - If you were using a plastic spoon instead of a metal spoon, how would the situation be different? - Give 1 example each of good conductor and bad conductor of heat. (or) - Why are cookware and pans not provided with metal handles?	[4] [1] [1] [1] [1] [1]
39.	A group of students were learning about the importance of vitamins and the role they play in maintaining good health. Help them reason out some of their learnings. - What disease does deficiency of Vitamin D cause? - What are water soluble vitamins? Why can't they be stored for a long time in the body? - What is the importance of vitamins in our diet? or - What are vitamins also called as?	[4] [1] [2] [1] [1]